DSEK-AE: A Dual-Stream Autoencoder Based on Endmember Kernel for Blind Hyperspectral Unmixing

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***Abstract*—Blind hyperspectral unmixing (HU) is an ill-posed inverse problem that poses significant challenges. Numerous model-based and deep-learning methods have been proposed to address it, achieving remarkable progress. However, these methods are predominantly based on the linear spectral mixing model, with insufficient exploration of nonlinear blind unmixing; furthermore, their results are highly sensitive to endmember initialization, and the use of intrinsic properties of hyperspectral images (HSIs) remains limited. To address these issues, this article proposes a dual-stream autoencoder based on endmember kernel (DSEK-AE), which enables blind linear/nonlinear unmixing through a knowledge-driven and data-driven approach. The encoder consists of two streams: the endmember extraction stream aggregates information from all pixels to directly derive endmembers, while the abundance estimation stream comprises three serial modules—spectral perception module (SPM), global-local aggregation module (GLAM), and sparsity adjustment layer (SAL)—which effectively capture both global and local spatial-spectral information from HSIs, significantly improving unmixing accuracy. The decoder is designed based on the endmember-kernel theory, explicitly modeling interactions among endmembers and enabling flexible adjustment of embedded spectral mixing models without altering the underlying network structure, thereby enhancing the network's physical interpretability and flexibility. Moreover, we propose a novel loss function specifically designed for blind unmixing to facilitate more effective training. Experimental results on real-world and physically simulated hyperspectral datasets consistently demonstrate that the proposed method exhibits superior unmixing performance compared to current state-of-the-art methods. The code will be available at https://github.com/I3ab/DSEK-AE**

***Index Terms*—Autoencoder (AE), dual-stream encoder, endmember-kernel-based decoder, hyperspectral images (HSIs), linear mixing model (LMM), nonlinear mixing models (NMMs), unmixing.**

I. Introduction

UNLIKE panchromatic and multispectral images, hyperspectral images (HSIs) boast a significantly enhanced spectral resolution, covering hundreds to thousands of bands ranging from ultraviolet to infrared, enabling the fine capture of spectral characteristics of materials and thus achieving precise classification and identification of ground objects [1]. Benefiting from this unique characteristic, HSIs have been widely applied in various fields [2], including mineral exploration and environmental monitoring. However, due to insufficient spatial resolution, individual pixels often exhibit mixed spectra of multiple components. The presence of mixed pixels poses challenges to the accuracy of HSI applications, requiring the unmixing process to extract the constituent materials (endmembers) within the pixels and their respective proportions (abundances).

Spectral unmixing can be broadly categorized into two types based on whether the endmembers are known in advance: supervised unmixing and unsupervised unmixing (i.e., blind unmixing) [3]. In the case of supervised unmixing, the endmembers are either known a priori or obtained through endmember extraction algorithms, and only the abundances need to be calculated. However, in practical applications, it is often impractical to obtain endmembers in advance, and the two-stage process of first extracting endmembers and then estimating abundances is prone to introducing additional errors, resulting in a significant decrease in unmixing accuracy. Therefore, blind unmixing—unsupervised methods that directly estimate endmembers and abundances from HSIs—is of greater importance for practical applications and is also more challenging. It can typically be divided into two main categories: model-based methods and deep-learning methods.

A. Model-Based Methods

In current model-based blind unmixing methods, the linear mixing model (LMM) is widely adopted due to its simplicity and clear physical significance [4]. The model assumes that the incident light interacts with only a single substance before reaching the sensor. VCA [5] and FCLS [6] are classic methods for endmember extraction and abundance estimation under the LMM assumption, respectively, and when used in series, they can simply achieve blind linear unmixing. However, the above methods, along with other approaches based on convex geometry theory, often face limitations due to the assumption of the existence of pure pixels, making them less applicable to highly mixed hyperspectral data. In addition to the above, there are other widely studied blind unmixing methods, such as those based on independent component analysis (ICA) [7], [8],

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which decompose the data into statistically independent source signals, enabling blind linear unmixing without the need for pure pixels; nonnegative matrix factorization (NMF) is another statistical method widely used in blind linear unmixing [9], [10], [11], favored for its high alignment with the LMM and the inherent nonnegativity constraints on endmembers and abundances.

However, in practical applications, the interaction of photons between different substances results in nonlinear mixing effects, typically manifested as multiple scattering and absorption of light [12]. In such cases, applying the LMM significantly reduces the accuracy of the unmixing results. Currently, various nonlinear mixing models (NMMs) have been proposed, such as the bilinear mixing models (BMMs) and the multilinear mixing (MLM) model [13]. Specifically, BMMs, including the generalized bilinear model (GBM) [14] and the polynomial postnonlinear model (PPNM) [15], can be used to describe second-order nonlinearity (i.e., secondary scattering), while the MLM model can be used to describe higher order nonlinearity (i.e., higher order scattering).

Based on different NMMs, model-based blind nonlinear unmixing methods can be constructed [16], which can be broadly categorized into two main types: geometric methods and statistical methods. Geometric methods primarily address the blind nonlinear unmixing problem through geometric projection techniques [17], [18], [19]; statistical methods, in contrast, leverage the statistical properties of hyperspectral data (such as independence, sparsity [20], or probability distributions) to separate mixed components. Although these methods provide effective solutions under certain assumptions, their performance usually depends on the accuracy of the assumed mixing models. However, accurately modeling real-world scenes in practical applications is quite challenging, leading to the emergence of various model-free approaches. Among these, kernel-based methods are particularly notable, which use a specific kernel function to map hyperspectral data to a high-dimensional space to achieve nonlinear unmixing, but this mapping usually lacks clear physical significance. To overcome this issue, K-Hype [21] introduced the endmember-kernel theory, using kernel functions to explicitly represent interactions among endmembers.

In general, model-based blind unmixing methods typically follow these steps: first, construct an objective function based on domain knowledge, and then use specific optimization algorithms or solving strategies to obtain the solution to the problem. However, in practical applications, constructing an objective function that meets actual requirements is quite challenging, and the optimal solutions to these complex mathematical models are often computationally intractable, thus resulting in low unmixing accuracy of these methods. Moreover, the reliance on iterative optimization processes for problem solving often results in excessively long processing times, making real-time applications even more difficult to achieve.

B. Deep-Learning Methods

In recent years, with the rapid development of artificial intelligence, many deep learning-based unmixing methods have emerged [22], [23]. In particular, due to the strong compatibility of the autoencoder (AE) architecture with the unmixing problem, many AE-based methods have been successfully applied in the field of unmixing [24], [25]. These methods typically use an encoder to extract features from HSIs to estimate abundances and then employ a decoder to reconstruct the HSIs from the estimated abundances. However, most of these early methods process pixels individually, failing to consider the spatial relationships among neighboring pixels [26], thus overlooking the rich spatial information in HSIs and limiting their unmixing accuracy in complex scenarios.

To overcome the above problem, various methods have been proposed to better use the spatial-spectral information of HSIs [27], [28], [29], [30]. 3D-NAE [31] employs 3-D convolution to extract features from HSIs to achieve joint learning of spectral and spatial information; however, it fails to capture global information within the HSIs. To obtain global features of the HSIs, DeepTrans [32] integrates convolutional AEs with transformers to capture long-range dependencies within the HSIs, thereby enhancing the network's feature extraction capabilities. To better extract the spatial-spectral features through a transformer-based architecture, unDAT [33] constructs a spatial feature extraction module and a spectral feature extraction module to make full use of the spatial correlation within homogeneous regions and the dependencies between adjacent spectral bands. To alleviate disruptions caused by patch-based tokenization, SRViT [34] is proposed and employs natural gradient descent to enhance endmember estimation stability under high-noise conditions. Because of the insufficient exploitation of spectral information in existing methods, MS^{2}AE-Net [35] proposes a joint spatial-spectral AE framework for hyperspectral unmixing (HU), systematically integrating multiscale features from both spectral and spatial domains. However, in the above methods, networks based on fully connected layers (FCLs) overlook spatial information, while CNN-based networks only use local spatial information; transformer-based networks, although capable of capturing global spatial information, often pay insufficient attention to local details. Besides, although existing networks such as the spatial-spectral attention bilateral network (SSABN) [36] and the transformer-CNN collaborative unmixing network (TCCU-Net) [37] combine convolution operations with transformer modules to jointly consider local and global information for comprehensive learning of spatial relationships, they are limited to solving unmixing problems based on the LMM. For nonlinear unmixing problems, few networks can effectively extract both global and local spatial-spectral information from HSIs simultaneously [38].

It is worth noting that many existing unmixing networks are primarily data-driven, lacking the incorporation of physical knowledge into the design of network structure and loss function [39]. For instance, EGU-Net [40] employs three convolutional layers to learn the nonlinear mixing process without incorporating any physical knowledge. Such methods learn mixing patterns directly from data through training, which not only requires a large amount of training data but also lacks physical interpretability, resulting in poor generalization performance and insufficient noise robustness. To

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solve this problem, Recent research has attempted to embed spectral mixing models into the design of network structure and loss function [41], aiming to achieve HU driven by both knowledge and data. For example, DTU-Net [42] proposes a multiscale dilated transformer network based on PPNM that can accurately characterize bilinear mixing effects, thereby addressing the second-order nonlinear mixing problem. HapkeCNN [43] leverages the Hapke model to achieve unmixing for intimate mixtures by designing an appropriate loss function to constrain nonlinear relationships in the hyperspectral data. MLM-3DAE [44] embeds the MLM model into the decoder structure, enabling effective handling of higher order nonlinear mixing problems. Similarly based on MLM, DAAN [45] explicitly models multiple scattering and incorporates homogeneous smoothing constraints to achieve more accurate nonlinear unmixing. EMLM-Net [46] employs deep unfolding techniques to construct the encoder and designs the decoder based on the extended MLM (EMLM) model [47], forming a dual-stream network to solve infinite-order and wavelength-dependent nonlinear mixing issues. To more accurately estimate the transition probabilities, DEAE [48] designs a diffusion-enhanced EMLM decoder, effectively improving the accuracy and robustness of nonlinear blind unmixing. However, all these methods are tailored to specific types of spectral mixing models, limiting their applicability and flexibility. When the application scenario changes (i.e., when the spectral mixing model varies), the network architecture needs to be redesigned, which not only increases the complexity of network design and implementation but also restricts the unmixing method's versatility and flexibility across different real-world scenarios.

In addition to the above issues, it is worth pointing out that in the current blind unmixing methods based on AE architecture, endmembers are typically obtained by extracting weights from specific layers of the decoder, making them highly sensitive to initialization. Recent research has begun to address this challenge by directly computing the endmembers. For example, A2SAN [49] obtains the endmembers by calculating the product of the feature map and the abundance map. DFFN [50] uses the abundance-reconstructed HSIs as input to directly compute endmembers through the network. SSAUN [51] achieves direct endmember estimation through an inversion strategy. MSGA-CD [52] classifies all pixels of the HSIs according to abundances and calculates endmembers through the weighted summation of pixels' spectra within each class. An unmixing framework that combines recurrent Kolmogorov–Arnold networks (KANs) with segment anything model (SAM) adapter [53] is proposed to reconstruct endmembers through specific linear combinations of spectral features. Although these attempts can reduce the dependence on initialization when solving for endmembers, they derive endmembers from processed information (such as abundances, partial input pixels, and pseudo-endmembers extracted from VCA), which may introduce noise and result in the loss of some critical information in the original HSIs. Furthermore, all these methods are based on the LMM and fail to fully leverage the characteristics of endmembers when designing the network.

C. Motivation and Contribution

In this article, to address the above issues, we propose a dual-stream AE based on endmember kernel (DSEK-AE) for blind linear/nonlinear HU. It consists of three main components: a dual-stream encoder, an endmember-kernel-based decoder, and a specially designed loss function. First, to mitigate the impact of endmember initialization, a dual-stream encoder is designed to separately compute endmembers and abundances. Given the distinct characteristics of endmembers and abundances (endmembers remain consistent across all pixels, while abundances vary with pixels), their estimation processes should also differ. Therefore, based on prior knowledge of endmembers and abundances (the spectral smoothness of endmembers and the spatial smoothness and sparsity of abundances), different structures are designed for each stream, where CNN and transformer are employed to fully exploit the spatial-spectral information of the input HSIs. Second, since the endmember-kernel theory can clearly describe the complex interactions among endmembers through kernel functions and can be used to flexibly express the LMM and various NMMs, we integrate the endmember-kernel theory into the decoder to fully leverage the advantages of deep-learning methods and the endmember-kernel theory. Finally, given the inherent complexity of the blind unmixing problem, particularly in the context of blind nonlinear unmixing, a loss function that regularizes both endmembers and nonlinear intensities is adopted to further enhance the network's unmixing performance.

The main contributions of this article can be summarized as follows.

1) We propose a novel AE framework for blind linear/nonlinear HU. The proposed framework incorporates physical knowledge into the network architecture design, integrating the physical interpretability of model-based methods with the powerful representation capability of deep-learning methods, thereby achieving high-precision blind HU.
2) A dual-stream encoder is designed to separately compute endmembers and abundances, thereby alleviating the significant impact of endmember initialization. Besides, a CNN-transformer hybrid architecture is constructed to fully exploit the spatial-spectral information of HSIs, and a learnable layer is introduced to adaptively regulate the spatial sparsity of abundances.
3) Unlike other unmixing networks designed specifically for particular spectral mixing models, the proposed method leverages endmember-kernel theory, enabling convenient switching between different spectral mixing models simply by replacing the kernel functions in the decoder, without altering the fundamental network architecture. This endows the proposed network with high flexibility and adaptability across multiple scenarios.
4) A novel composite loss function is designed for blind HU. By incorporating reconstruction loss and regularizations on endmembers and nonlinear intensities into the loss function, the training of the network can be more effectively guided, thereby comprehensively enhancing the unmixing performance.

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The remainder of this article is organized as follows. Section II provides a brief review of the spectral mixing models and endmember-kernel theory. Section III elaborates on the proposed method. In Section IV, the performance of the proposed method is evaluated on multiple datasets, and the experimental results are thoroughly analyzed. Finally, the conclusion is presented in Section V.

II. Related Works

A. Spectral Mixing Models

Consider a 2-D matrix $\boldsymbol{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N\} \in \mathbb{R}^{B\times N}$ with B bands and N pixels obtained by rearranging the hyperspectral remote sensing imagery $\mathcal{X} \in \mathbb{R}^{B\times H\times W}$ (3-D tensor) with a spatial size of $H \times W$ pixels in a pixel-wise manner. Let $\boldsymbol{x_j}$ represent the jth pixel spectrum of $\boldsymbol{X}$, $\boldsymbol{m_i}$ denote the ith endmember in the endmember matrix $\boldsymbol{M} = \{\boldsymbol{m}_1, \boldsymbol{m}_2, \ldots, \boldsymbol{m}_R\} \in \mathbb{R}^{B\times R}$ whose (i, j)th entry is $m_{i,j}$, and R be the number of endmembers. $\boldsymbol{A} = \{\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_N\} \in \mathbb{R}^{R\times N}$ is the abundance matrix and its element $a_{i,j}$ is the abundance of the ith endmember for the jth pixel. $\boldsymbol{e_j}$ denotes the noise in the jth pixel. Then, a general spectral mixing model can be written as follows [1]:

$$\boldsymbol{x_j} = \mathcal{F}(\boldsymbol{M}, \boldsymbol{A}) + \boldsymbol{e_j} \tag{1}$$

where $\mathcal{F}(\cdot)$ is a function characterizing the intrinsic interaction mechanism of photons.

The LMM [1] posits that each pixel spectrum is a linear combination of endmembers. Then the LMM can be formulated as follows:

$$\boldsymbol{x_j} = \sum_{i=1}^{R} \boldsymbol{m}_i a_{i,j} + \boldsymbol{e_j}. \tag{2}$$

Considering the physical meanings of endmembers and abundances, the endmembers should satisfy the endmember nonnegativity constraint (ENC), while the abundances need to meet both the abundance nonnegativity constraint (ANC) and the abundance sum-to-one constraint (ASC), as described in the following equation:

$$m_{i,j} \geq 0, \quad a_{i,j} \geq 0, \quad \sum_{i=1}^{R} a_{i,j} = 1. \tag{3}$$

The BMM extends the LMM by introducing a bilinear term to simulate second-order scattering effects of light. Commonly used BMMs primarily include the GBM [14] and PPNM [15].

The GBM [14] can be given as follows:

$$\boldsymbol{x_j} = \sum_{i=1}^{R} \boldsymbol{m}_i a_{i,j} + \sum_{i=1}^{R-1} \sum_{k=i+1}^{R} \gamma_{i,j,k} (\boldsymbol{m_i} \odot \boldsymbol{m_k}) a_{i,j} a_{k,j} + \boldsymbol{e_j} \tag{4}$$

where $\gamma_{i,j,k} \in [0, 1]$ is a parameter measuring the nonlinear intensity and $\odot$ is the Hadamard elementwise product. Here, the constraints on the endmembers and abundances are also employed.

Compared to the GBM, the PPNM [15] takes into account the interactions between the same endmembers. Under the constraints of the ENC, ANC, and ASC, it can be expressed as follows:

$$\boldsymbol{x_j} = \sum_{i=1}^{R} \boldsymbol{m}_i a_{i,j} + b_j \sum_{i=1}^{R} \sum_{k=1}^{R} (\boldsymbol{m_i} \odot \boldsymbol{m_k}) a_{i,j} a_{k,j} + \boldsymbol{e_j} \tag{5}$$

where $b_j \in \mathbb{R}$ serves to control the contribution of the nonlinear term.

Higher order NMMs further involve higher order scatterings among endmembers. A typical model is MLM [13], and it has the following mathematical expression:

$$\begin{aligned} \boldsymbol{x_j} = {} & (1 - p_j) \sum_{i=1}^{R} \boldsymbol{m}_i a_{i,j} \\ & + (1 - p_j) p_j \sum_{i=1}^{R} \sum_{k=1}^{R} (\boldsymbol{m_i} \odot \boldsymbol{m_k}) a_{i,j} a_{k,j} \\ & + (1 - p_j) p_j^2 \sum_{i=1}^{R} \sum_{k=1}^{R} \sum_{l=1}^{R} (\boldsymbol{m_i} \odot \boldsymbol{m_k} \odot \boldsymbol{m_l}) a_{i,j} a_{k,j} a_{l,j} \\ & + \cdots + \boldsymbol{e_j} \end{aligned} \tag{6}$$

where p_j is the probability that light further interacts with materials, constrained to $p_j < 1$. Besides, the endmembers and abundances are also subject to the ENC, ANC, and ASC, respectively.

B. Endmember-Kernel Theory

Typically, kernel-based methods map hyperspectral data into a high-dimensional space and replace inner products with kernel functions when optimizing the objective function, thereby transforming nonlinear problems into linear ones for easier solution; however, these methods usually lack clear physical interpretations. In contrast to the above approaches, K-Hype [21] introduces endmember-kernel theory, explicitly modeling nonlinear interactions among endmembers using kernel functions and providing a more physically meaningful interpretation [54].

Specifically, the endmember-kernel theory [21] assumes the pixel spectrum can be formulated as the combination of a linear component and additive nonlinear fluctuations defined in reproducing kernel Hilbert spaces [21]

$$\boldsymbol{x}_j = \boldsymbol{M}\boldsymbol{a}_j + \Phi(\boldsymbol{M}) + \boldsymbol{e}_j \tag{7}$$

where Φ is an unknown function describing the nonlinear interactions among endmembers. Given that the derivation of the endmember-kernel theory is based on such a universal mixing mechanism, this approach is applicable to diverse mixing scenarios.

Based on (7), a least squares support vector machine (LS-SVM) problem is formulated [21]

$$\begin{aligned} < \boldsymbol{a}_j^*, \varphi_j^* > = {} & \arg\min_{\boldsymbol{a}_j,\, \varphi_j} \frac{1}{2} \left(\left\| \boldsymbol{a}_j \right\|^2 + \left\| \varphi_j \right\|_{\mathcal{H}}^2 + \frac{1}{\mu} \left\| \boldsymbol{e}_j \right\|^2 \right) \\ & s.t.\ \boldsymbol{a}_j \geq 0, \quad \boldsymbol{1}^T \boldsymbol{a}_j = 1 \\ & e_{l,j} = x_{l,j} - \boldsymbol{m}_{\lambda_l} \boldsymbol{a}_j - \varphi_j(\boldsymbol{m}_{\lambda_l}) \\ & l = 1, \ldots, B \end{aligned} \tag{8}$$

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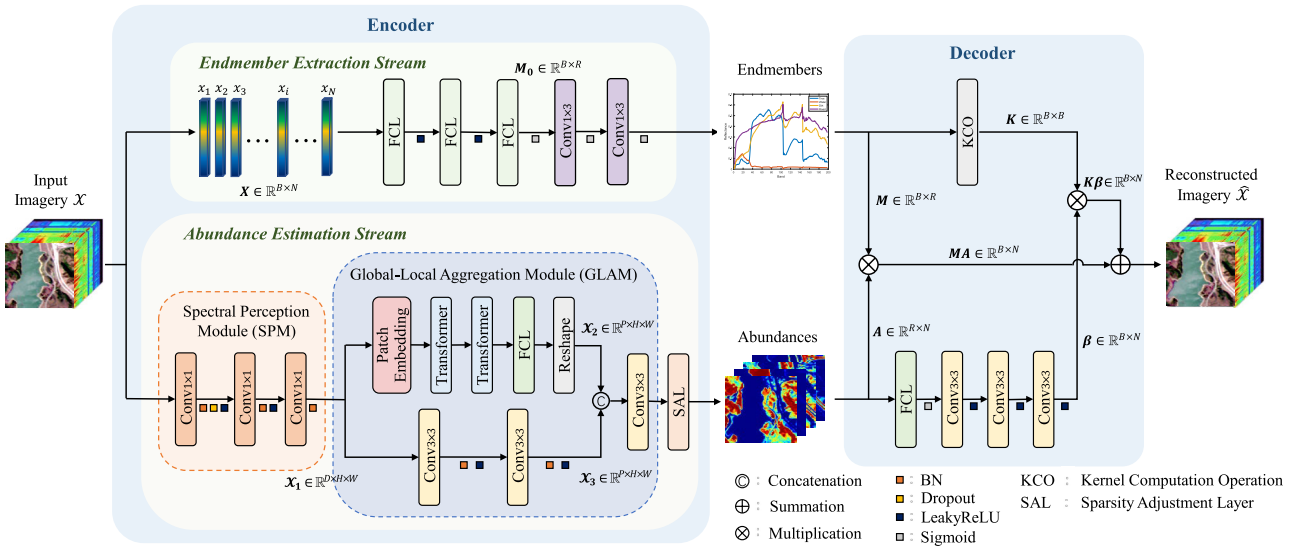


Fig. 1. Overall framework of DSEK-AE.

where $\boldsymbol{m}_{\lambda_l} \in \mathbb{R}^{1\times R}$ denotes the lth band of the endmember matrix. By applying duality theory [55] and the theory of reproducing kernels [56] to the above constrained optimization problem, we have

$$\begin{cases} \boldsymbol{a}_j^* = \displaystyle\sum_{l=1}^{B} \beta_{l,j}^* \boldsymbol{m}_{\lambda_l} + \boldsymbol{\tau}_j^* - \theta_j^* \mathbf{1} \\ \varphi_j^* = \displaystyle\sum_{l=1}^{B} \beta_{l,j}^* \kappa\left(\cdot, \boldsymbol{m}_{\lambda_l}\right) \\ e_{l,j}^* = \mu \beta_{l,j}^* \end{cases} \tag{9}$$

where κ refers to reproducing kernel and $\beta_{l,j}$, $\boldsymbol{\tau}_j$, and θ_j are the Lagrange multipliers.

III. Proposed Method

To address the above issues, we propose DSEK-AE for blind HU, which adopts an AE architecture. The encoder consists of two streams: the endmember extraction stream employs FCLs and CNNs to directly derive endmembers, while the abundance estimation stream comprises three serial modules—spectral perception module (SPM), global-local aggregation module (GLAM), and sparsity adjustment layer (SAL)—to estimate abundances more accurately. The decoder is based on endmember-kernel theory to model the spectral mixing process and output the reconstructed imagery. The overall framework of the proposed DSEK-AE is shown in Fig. 1. Besides, the loss function is specially designed for blind HU to better guide the training.

With adequate training, the network is able to generate precise unmixing results. The following is a detailed description of each part of the proposed method.

A. Encoder

1) Endmember Extraction Stream: Rather than using the weights of a certain layer of the decoder as endmembers, the endmember extraction stream directly computes endmembers from the input HSIs, thereby alleviating the influence of endmember initialization.

First, to facilitate subsequent processing, the input imagery $\mathcal{X} \in \mathbb{R}^{B\times H\times W}$ is reshaped into a 2-D matrix $\boldsymbol{X} \in \mathbb{R}^{B\times N}$ ($N = H\times W$). Since all pixels share the same set of endmember spectra, three FCLs with $R \times 200$, $R \times 50$, and R nodes, respectively are employed to fully extract the information from the input. The output of this process is $\boldsymbol{M}_0 \in \mathbb{R}^{B\times R}$. To introduce nonlinearity into the network, the LeakyReLU is used as the activation function in the first two layers, while sigmoid is applied in the third layer. Then, considering the inherent smoothness across adjacent bands in HSIs, 1-D convolutional layers, each followed by a sigmoid activation function, are introduced to better promote spectral smoothness of the extracted endmembers. Two such layers are used to enlarge the receptive field and enhance modeling capability. Here, the original intention of using the sigmoid function was to enforce nonnegativity and boundedness, while numerical stability is an additional beneficial effect. This process can be mathematically expressed as follows:

$$\hat{\boldsymbol{M}} = \text{Sigmoid}\left(\text{Conv}_{1\times 3}\left(\text{Sigmoid}\left(\text{Conv}_{1\times 3}\left(\boldsymbol{M}_0\right)\right)\right)\right) \tag{10}$$

where $\hat{\boldsymbol{M}} \in \mathbb{R}^{B\times R}$ represents the extracted endmembers.

It should be noted that the dimension selection of components is determined by the physical meaning of the input and output, and in the absence of strict requirements, it is based on the trade-off between model size and representational capacity.

2) Abundance Estimation Stream: The abundance estimation stream is composed of SPM, GLAM, and SAL, aiming to extract the spatial-spectral information of the input HSIs and derive the corresponding abundances.

1) *SPM:* In SPM, the input imagery is first processed through three 1×1 convolutional layers, each followed by a batch normalization (BN). 1×1 convolutional layer does not involve spatial relationships, which enables the exclusive analysis of spectral features while preserving the original spatial information. Moreover, this

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process effectively reduces the dimensionality of the high-dimensional HSIs, eliminating spectral redundancy and thereby reducing the computational complexity of subsequent processing. Besides, to prevent overfitting, dropout is applied in the first layer, and the first two layers employ the LeakyReLU activation function to introduce nonlinearity. Specifically, given a hyperspectral imagery $\mathcal{X} \in \mathbb{R}^{B\times H\times W}$, the output of SPM is a low-dimensional representation $\mathcal{X}_1 \in \mathbb{R}^{D\times H\times W}$ $(D \ll B)$ that preserves the same spatial dimensions.

2) *GLAM:* Considering that CNNs excel at extracting local features but perform poorly in capturing long-range dependencies, while transformers effectively capture global dependencies yet often struggle to preserve local details, GLAM adopts a parallel structure by combining CNN and transformer to fully leverage the strengths of each component, thereby enhancing the network's feature extraction capability. To be specific, the entire GLAM is a dual-branch network. The upper branch consists of two transformer blocks, which are capable of extracting long-range dependencies, while the lower branch includes two 3×3 convolutional layers, each followed by a BN and a LeakyReLU activation function, to extract local information. To facilitate the integration of information from two branches, an FCL and a reshaping operation are used to convert the upper branch's output to the shape of the lower branch. Then the outputs of the two branches are concatenated together and the features from two branches are fused by a 3×3 convolutional layer. The output of GLAM can be described as follows:

$$\mathcal{P} = \mathrm{Conv}_{3\times 3}\left(\mathrm{Concat}\left(\mathrm{Branch}_{\mathrm{upper}}\left(\mathcal{X}_1\right), \mathrm{Branch}_{\mathrm{lower}}\left(\mathcal{X}_1\right)\right)\right). \tag{11}$$

3) *SAL:* To satisfy the ANC and ASC, SAL is designed with the following mathematical expression:

$$\hat{a}_{i,j} = \frac{\exp\left(T \times p_{i,j}\right)}{\sum_{i=1}^{R} \exp\left(T \times p_{i,j}\right)} \tag{12}$$

where $P \in \mathbb{R}^{R\times N}$ is the matrix obtained by rearranging the output of GLAM $\mathcal{P} \in \mathbb{R}^{R\times H\times W}$ along the pixel dimension, and $p_{i,j}$ is the element at the ith row and the jth column of P. In fact, SAL can be viewed as an extension of softmax with an additional data-dependent scalar parameter $T \in (0, +\infty)$, and the original softmax function only corresponds to the special case where $T = 1$. SAL allows for adaptively regulating the sparsity of abundances by adjusting the value of T. Specifically, as T increases, SAL makes the data distribution exhibit a steeper profile, potentially causing more values to approach zero, which implies a higher degree of abundance sparsity; conversely, the opposite occurs. Since T varies with the input imagery, it is designed as a learnable parameter. Through training, T is optimized from data, thereby enabling adaptive adjustment of abundances' sparsity. Regarding sparsity imposition, although L_0 regularization theoretically achieves strict sparsity, it is often replaced in practice with L_1 regularization due to its more tractable optimization. However, L_1 regularization still suffers from poor training stability and requires additional constraints to ensure the nonnegativity and sum-to-one properties of abundances. In contrast, SAL effectively achieves sparsity regularization, enforces physical constraints on abundances, and ensures training stability, while effectively balancing all three aspects, making it more practical.

In summary, the input imagery undergoes processing sequentially through SPM, GLAM, and SAL, ultimately yielding abundance information.

B. Decoder

By substituting (9) into (7), a unified description framework for different mixing scenarios can be given as follows:

$$\boldsymbol{X} = \boldsymbol{M}\boldsymbol{A} + \boldsymbol{K}\boldsymbol{\beta} \tag{13}$$

where $\boldsymbol{K} \in \mathbb{R}^{B\times B}$ is the Gram matrix, also known as the endmember-kernel matrix, and $\boldsymbol{\beta} \in \mathbb{R}^{B\times N}$ is used to characterize nonlinear intensities (i.e., the magnitudes of the nonlinear contributions), which can be intuitively regarded as a comprehensive expression composed of abundances and nonlinear parameters. The form of the endmember-kernel matrix $\boldsymbol{K}$ reflects the mixing relationships among endmembers and can be used to characterize different types of spectral mixing (or correspondingly represent various types of land-cover scenarios), while the nonlinear intensities $\boldsymbol{\beta}$ adjust the strength of nonlinearity. For different mixing scenarios, $\boldsymbol{K}$ varies. Evidently, when $\boldsymbol{K}$ is set as $\mathbf{0}$, it is equivalent to the LMM. Therefore, the endmember-kernel matrix of the LMM can be written as follows:

$$\boldsymbol{K}_{LMM} = \mathbf{0}. \tag{14}$$

The BMM can be used to describe and elucidate the second-order scattering effects of light, primarily encompassing the GBM [14] and the PPNM [15].

For the GBM scenario that does not account for interactions between identical endmembers, the corresponding expression of $\boldsymbol{K}$ is shown as follows [57]:

$$\boldsymbol{K}_{GBM} = \left(\boldsymbol{M}\boldsymbol{M}^T\right)^2 - \boldsymbol{M}^2\boldsymbol{M}^{2T} \tag{15}$$

where $(\cdot)^2$ denotes the element-wise squaring of the matrix.

For the PPNM scenario considering interactions between identical endmembers, the corresponding endmember-kernel matrix expression is as follows [57]:

$$\boldsymbol{K}_{PPNM} = \left(\boldsymbol{M}\boldsymbol{M}^T\right)^2. \tag{16}$$

For the MLM scenario capable of describing and explaining infinite-order mixing processes, the corresponding endmember-kernel matrix can be expressed through the Gaussian kernel function [54]

$$\kappa_{MLM}\left(\boldsymbol{m}_{\lambda_i}, \boldsymbol{m}_{\lambda_j}\right) = \exp\left(-\frac{\left|\left|\boldsymbol{m}_{\lambda_i} - \boldsymbol{m}_{\lambda_j}\right|\right|^2}{2\sigma^2}\right) \tag{17}$$

where σ is a parameter and in the following experiment, σ is set to 1.

Based on (13), the decoder can model the spectral mixing process according to the endmember-kernel theory, and it

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consists of two parts: a linear part and a nonlinear part. Therefore, the decoder is designed with two corresponding branches: a linear branch and a nonlinear branch.

The linear branch conducts the linear reconstruction of the imagery by computing the product of endmembers and abundances, which are produced from the output of the encoder.

In the nonlinear branch, kernel computation operation (KCO) calculates $\boldsymbol{K}$ by substituting the estimated endmembers into the corresponding formulas from (14) to (17). As can be seen from (9), $\boldsymbol{\beta}$ is used to characterize nonlinear intensities and can be understood as a comprehensive manifestation of abundances and nonlinear parameters. Therefore, an FCL followed by a sigmoid activation function and three convolutional layers, each followed by a LeakyReLU activation function, are employed to fit this functional relationship. Then the nonlinear reconstruction of the imagery is obtained by multiplying $\boldsymbol{K}$ with $\boldsymbol{\beta}$. The rationale behind using three consecutive convolutional layers is to expand the receptive field, capture broader spatial contextual information, and enhance the network's representational capacity.

Finally, the outputs of the linear branch and the nonlinear branch are summed to obtain the reconstructed imagery.

Because of the integration of specific spectral mixing models within the decoder, the network's physical interpretability is significantly enhanced. Furthermore, since different kernel matrices correspond to different spectral mixing models (including LMM and various NMMs), adapting to different scenarios does not require redesigning the network architecture for different spectral mixing models. For practical applications, with the help of the known prior knowledge of the mixing type of landcovers in HSIs, the switching between different spectral mixing models can be easily realized without changing the basic structure of the proposed unmixing network by simply replacing the kernel matrix embedded in the decoder.

C. Loss Function

The loss function contains three components: reconstruction loss, as well as regularization terms of the endmembers $\boldsymbol{M}$ and nonlinear intensities $\boldsymbol{\beta}$.

To better measure the similarity between the input imagery and the reconstructed imagery, we combine spectral similarity metrics based on spectral angle and Euclidean distance.

Let $\boldsymbol{x}_j$ and $\hat{\boldsymbol{x}}_j$ represent the spectrum of the jth pixel of the input imagery $\boldsymbol{X}$ and the reconstructed imagery $\hat{\boldsymbol{X}}$, respectively, and $x_{i,j}$ and $\hat{x}_{i,j}$ represent the element located at the ith row and the jth column of the input imagery $\boldsymbol{X}$ and the reconstructed imagery $\hat{\boldsymbol{X}}$, respectively. L_{SAM}, which quantifies spectral similarity by assessing angular differences between high-dimensional vectors, reflects discrepancies in spectral shape, and its definition can be stated as follows:

$$L_{SAM} = \frac{1}{N}\sum_{j=1}^{N} \arccos\left(\frac{\boldsymbol{x}_j^T \hat{\boldsymbol{x}}_j}{\|\boldsymbol{x}_j\|_2 \|\hat{\boldsymbol{x}}_j\|_2}\right). \tag{18}$$

On the other hand, L_{ED} measures reconstruction error based on Euclidean distance, and its definition can be described as follows:

$$L_{ED} = \frac{1}{BN}\sum_{i=1}^{B}\sum_{j=1}^{N}\left(\hat{x}_{i,j} - x_{i,j}\right)^2. \tag{19}$$

Given the complexity of the blind HU problem and its vast solution space, solving it without appropriate regularizations may result in physically implausible solutions or convergence to suboptimal local minima. Therefore, we impose regularizations on $\boldsymbol{M}$ and $\boldsymbol{\beta}$, respectively.

To address distinct optimization requirements at different training stages, we design a two-stage loss function, where the only difference between the two stages lies in the regularization applied to the endmembers.

Specifically, in the first stage, the results from other simple algorithms (such as VCA) are used to guide the network in learning endmembers. This guidance does not need to be highly accurate because the purpose of this is not to provide precise supervision, but rather to offer a rough, physically meaningful direction for early endmember learning

$$L_M = \frac{1}{R}\sum_{i=1}^{R} \arccos\left(\frac{\hat{\boldsymbol{m}}_i^T \tilde{\boldsymbol{m}}_i}{\|\hat{\boldsymbol{m}}_i\|_2 \|\tilde{\boldsymbol{m}}_i\|_2}\right) \tag{20}$$

where $\hat{\boldsymbol{m}}_i$ and $\tilde{\boldsymbol{m}}_i$ represent the ith endmember in the estimated endmember matrix $\hat{\boldsymbol{M}}$ and the guidance endmember matrix $\tilde{\boldsymbol{M}}$, respectively.

In the second stage, the regularization strategy is adjusted to prevent the estimated endmembers from being excessively constrained by the early guidance, while also avoiding unreasonable situations where the estimated endmembers deviate significantly from the distribution of observed data due to insufficient constraints

$$L_M = \frac{1}{BR}\sum_{i=1}^{R} \left\|\hat{\boldsymbol{m}}_i - \bar{\boldsymbol{X}}\right\|_2^2 \tag{21}$$

where $\bar{\boldsymbol{X}}$ represents the average spectrum of the input. It should be noted that this term (L_M) serves only as a regularization term for endmembers, while the ultimately extracted endmembers are determined jointly by the reconstruction losses and all regularization terms.

L_N is a regularization term of nonlinear intensities $\boldsymbol{\beta}$, which is specifically designed to handle the inherent complexity in the blind nonlinear HU problem

$$L_N = \|\boldsymbol{\beta}\|_2 . \tag{22}$$

To sum up, the total loss function can be expressed as follows:

$$L_{total} = L_{SAM} + \lambda_{ED} L_{ED} + \lambda_M L_M + \lambda_N L_N \tag{23}$$

where λ_{ED}, λ_M, and λ_N are the tradeoff parameters that balance the contribution of each term within the loss function.

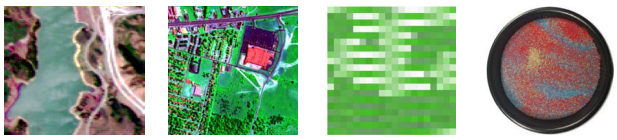
The pseudocode of the proposed DSEK-AE is summarized in Algorithm 1. Note that the endmember extraction stream, the SPM, and the process of calculating $\boldsymbol{\beta}$ are represented by $f_{EES}(\cdot)$, $f_{SPM}(\cdot)$, and $f_{cal\beta}(\cdot)$, respectively.

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Algorithm 1 Pseudocode of the Proposed DSEK-AE

Input: HSI $\mathcal{X} \in \mathbb{R}^{B\times H\times W}$.
Output: Endmembers $\boldsymbol{M} \in \mathbb{R}^{B\times R}$, Abundances $\boldsymbol{A} \in \mathbb{R}^{R\times N}$.

```
Obtain the guidance endmembers M̃ by VCA.
while stopping conditions are not met, do
    % Encoder
    Calculate the endmembers M = f_EES(X);
    Calculate X_1 = f_SPM(X);
    Calculate P by (11);
    Calculate the abundances A by (12);
    % Decoder
    Calculate K by the applicable equation among (14)–(17);
    Calculate β = f_calβ(A);
    Calculate the reconstructed imagery by (13);
    % Loss Function
    if first stage
    Calculate L_M by (20);
    else
    Calculate L_M by (21);
    end
    Calculate the total loss L_total by (23);
end
```



(a) (b) (c) (d)

Fig. 2. Hyperspectral datasets used in the experiments. (a) Jasper Ridge, (b) Urban, (c) Orchard, and (d) Laboratory-created datasets.

IV. Experiments

In this section, experiments are conducted on four datasets with diverse characteristics to comprehensively validate the performance of the proposed method. First, we introduce the datasets and the evaluation metrics used in the experiments. Then, the experimental results of the proposed method on various datasets are presented, along with comparisons to other algorithms. Finally, further experiments are conducted to deeply analyze and assess the proposed method.

A. Dataset Description

To comprehensively evaluate the performance of the proposed method, four hyperspectral datasets with distinct mixing characteristics across different scenarios are employed in the experiments, as illustrated in Fig. 2. The Jasper Ridge dataset [58] and the Urban dataset [59] are two commonly used real-world hyperspectral datasets representing typical natural landscape and urban scene characteristics, respectively. The Orchard dataset [60] is a semi-physical, semi-synthetic dataset, while the laboratory-created dataset [61] was acquired by imaging mixed quartz sand scenes in a laboratory setting using a hyperspectral camera. Therefore, for the latter two datasets, accurate endmember spectra and their abundance information are available, which we refer to as "ground truth" in subsequent experiments. Accordingly, for the first two real-world datasets that do not contain ground truth labels, to maintain rigor, we adopt the term "ground reference."

1) Jasper Ridge Dataset: The Jasper Ridge dataset [58] was collected by the Jet Propulsion Laboratory (JPL) using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), covering the Jasper Ridge Biological Preserve in California, as shown in Fig. 2(a). The full image size is 512×614 pixels, consisting of 224 spectral bands spanning the range of 380–2500 nm. For unmixing studies, a commonly used 100 × 100-pixel subimage is extracted, with 26 heavily affected bands due to atmospheric and water vapor interference removed, leaving 198 bands. Water, road, dirt, and tree are the representative land-cover types in the subimage.

2) Urban Dataset: The Urban dataset [59], captured by the Hyperspectral Digital Image Collection Experiment (HYDICE), describes an urban area at Copperas Cove, U.S., as shown in Fig. 2(b). It is a relatively large-scale dataset among the currently publicly available hyperspectral datasets with ground reference information, which contains 307 × 307 pixels and has 210 bands covering from 400 to 2500 nm. After removing the corrupted bands 1–4, 76, 87, 101–111, 136–153, and 198–210, 162 bands remain. In the experiment, a 300 × 300-pixel cropped image is used, and there are four endmembers in this scene: asphalt, grass, tree, and roof.

3) Orchard Dataset: The Orchard dataset [60] was generated through simulations using an extended physically based ray tracer (PBRT), as shown in Fig. 2(c). Its light sources, sensor platform, physical optical properties, and the geometric description were designed to closely approximate real-world conditions, allowing for a realistic simulation of higher order nonlinear mixing phenomena. The dataset consists of 20 × 20 pixels with a spatial resolution of 2 m, covering 216 bands from 350 to 2500 nm. After removing contaminated bands, 185 bands remain. The entire scene is composed of three endmembers: tree, soil, and weed.

4) Laboratory-Created Dataset: The laboratory-created dataset [61] simulates intimate mixing scenarios by mixing three differently colored types of quartz sand in the laboratory, as shown in Fig. 2(d). The dataset contains 256 bands, covering the visible to near-infrared wavelengths ranging from 400 to 1000 nm. The central region of 60×60 pixels is cropped for the experiment, with different colored quartz sands treated as endmembers. In this dataset, the percentage of each color of sand is considered to be the abundance, which can be evaluated by analyzing the associated high-resolution RGB image.

B. Evaluation Metrics

In line with conventional practices, we use average spectral angle distance (aSAD) to evaluate the results of endmember extraction and average root mean square error (aRMSE) to assess the performance of abundance estimation. The formulas for calculating aSAD and aRMSE are as follows:

$$aSAD = \frac{1}{R}\sum_{i=1}^{R}\arccos\left(\frac{\boldsymbol{m}_i^T\hat{\boldsymbol{m}}_i}{\|\boldsymbol{m}_i\|_2\ \|\hat{\boldsymbol{m}}_i\|_2}\right) \tag{24}$$

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TABLE I
HYPERPARAMETER SETTINGS FOR THE JASPER RIDGE DATASET

Methods	Details of Hyperparameter Settings
A2SAN [49]	$learning\ rate = 5\times10^{-2},\ epoch = 1000,\ \beta = 10^{-4}$
UNMamba [29]	$learning\ rate = 3\times10^{-3},\ epoch = 1000,\ \alpha = 1,\ \beta = 10^{-4}$
3D-NAE [31]	$learning\ rate = 10^{-3},\ epoch = 200,\ \lambda = 10^{-3},\ \gamma = 10^{-6}$
DTU-Net [42]	$learning\ rate = 4.5\times10^{-2}\ (10^{-2}\ for\ linear\ part),\ epoch = 600,\ C = 72,\ \alpha = 0.5, \gamma = 1$
HapkeCNN [43]	$learning\ rate = 10^{-3},\ epoch = 1000,\ \alpha = 0.1,\ \lambda = 0.15$
MLM-3DAE [44]	$learning\ rate = 10^{-4},\ epoch = 300$
DSEK-AE (Ours)	$learning\ rate = 5\times10^{-3},\ epoch = 240,\ \lambda_{ED} = 10^{-2},\ \lambda_{M} = 1,\ \lambda_{N} = 10^{-1}$

$$aRMSE = \sqrt{\frac{1}{RN}\sum_{j=1}^{N}\left\|\boldsymbol{a}_j - \hat{\boldsymbol{a}}_j\right\|_2^2} \tag{25}$$

where $\hat{\boldsymbol{m}}_i$ and $\hat{\boldsymbol{a}}_j$ are the ith estimated endmember and the estimated abundance for the jth pixel, respectively.

C. Comparison With Other Methods

To assess the proposed method, several advanced unmixing approaches are used for comparison, involving three linear unmixing methods (VCA + FCLS [5], [6], A2SAN [49], and UNMamba [29]) and four nonlinear unmixing methods (3D-NAE [31], DTU-Net [42], HapkeCNN [43], and MLM-3DAE [44]). VCA and FCLS (V/F) are classical methods for endmember extraction and abundance estimation, respectively. A2SAN and UNMamba are currently advanced linear blind unmixing algorithms, and they directly calculate endmembers instead of deriving them from network weights, which significantly reduces the impact of endmember initialization. The four nonlinear methods are suitable for different spectral mixing scenarios. Specifically, 3D-NAE relies on a neural network to learn the nonlinear mixing mechanism from the data. DTU-Net is primarily designed for bilinear unmixing problems. HapkeCNN constructs a Hapke model-based AE network for the unmixing of intimate mixtures. MLM-3DAE is developed based on the MLM to address nonlinear HU problems involving higher order scatterings.

For the comparison algorithms used in the experiments, the codes are obtained from their corresponding papers [5], [6], [29], [31], [42], [43], [44], [49], with the hyperparameter settings following suggestions provided in these papers. Taking the Jasper Ridge dataset as an example, Table I lists the corresponding key hyperparameter values for each algorithm. To evaluate the performance stability and reliability, all methods are run multiple times, and the mean and standard deviation of aSAD and aRMSE are reported.

It should be noted that since all deep learning-based comparative methods, as well as our method, are based on AE architectures and fall under the category of self-supervised learning, the training procedures are conducted on the entire hyperspectral imagery. Furthermore, for our method, the two-stage training procedure is essentially divided based on the total number of training epochs: specifically, the first stage comprises the initial one-third of the total epochs, while the second stage covers the remaining two-thirds.

1) Experiments on the Jasper Ridge Dataset: Table II presents the results of endmember extraction and abundance estimation obtained from different methods applied to the Jasper Ridge dataset. Among the compared methods, UNMamba performs relatively well due to its full use of image information. Compared to all these methods, the proposed methods yield the best results. The Jasper Ridge dataset primarily consists of water, road, dirt, and tree. The significant height differences between the tree endmember and other endmembers lead to second-order scattering being dominant. Besides, strong scattering also exists between water endmembers themselves. Therefore, the proposed method with PPNM kernel performs better because it more accurately reflects the physical characteristics of the Jasper Ridge dataset. Figs. 3 and 4 present the visualization results of abundance estimation and endmember extraction, respectively. It is evident that the proposed method with the PPNM kernel exhibits the closest proximity to the ground reference.

2) Experiments on the Urban Dataset: The results on the Urban dataset further substantiate the effectiveness of the proposed method. The Urban dataset exhibits distinct characteristics of urban scenes and is more complex compared to the Jasper Ridge dataset. Moreover, certain endmembers present high spectral similarity within specific bands [59] (i.e., the spectra of grass and tree both exhibit typical vegetation characteristics, with pronounced chlorophyll absorption in the visible light region and a sharp increase in reflectance in the near-infrared region, making them difficult to distinguish), thus increasing the unmixing difficulty. Table III and Figs. 5 and 6 present the evaluation results of different algorithms on the Urban dataset from both quantitative and qualitative perspectives, respectively. As shown in Table III, due to the complexity of unmixing the Urban dataset, most algorithms perform poorly on this dataset. In contrast, our methods, benefiting from the incorporation of prior information and the effective use of spatial-spectral features, demonstrate significant improvements in both abundance estimation and endmember extraction compared to other methods. Figs. 5 and 6 provide a more intuitive illustration of the advantages of the proposed method, which show that the endmembers and abundances obtained by the proposed DSEK-AE are in the highest consistency with the ground reference.

3) Experiments on the Orchard Dataset: The quantitative results of different methods on the Orchard dataset are presented in Table IV. The proposed method with the MLM kernel achieves the best performance. According to the scene generation process, the Orchard dataset involves significant multiple scattering phenomena [60]; MLM can effectively characterize higher order interactions caused by multiple reflections, making it suitable for scenarios with complex geometries such as vegetation canopies [13], which aligns well with the characteristics of the Orchard dataset. Therefore, it is reasonable that the methods based on LMM (V/F, UNMamba, A2SAN, and DSEK-AE with LMM kernel) and BMMs (DTU-Net, DSEK-AE with GBM kernel, and DSEK-AE with PPNM kernel) perform poorly, whereas the methods based on the MLM (MLM-3DAE and DSEK-AE with MLM kernel) yield better results. Although HapkeCNN accounts for higher

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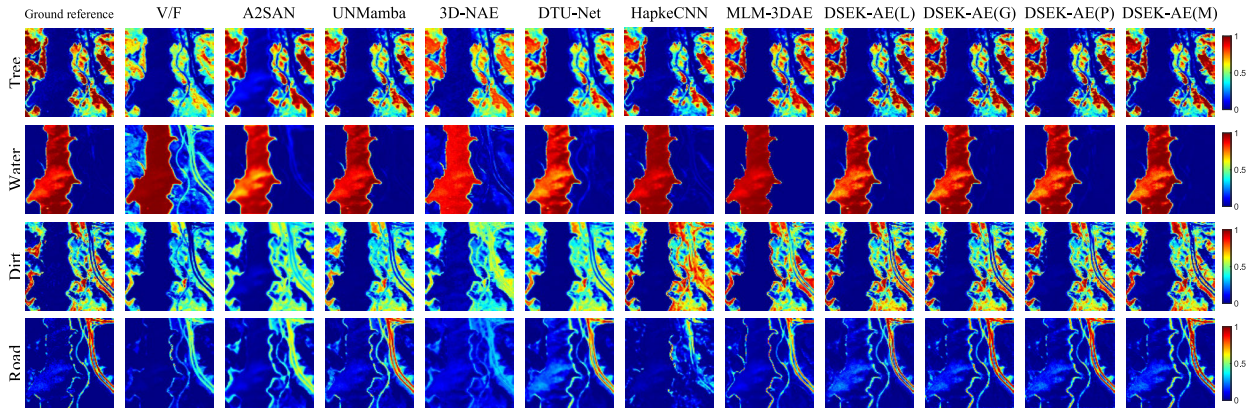


Fig. 3. Abundance maps obtained by different methods for the Jasper Ridge dataset.

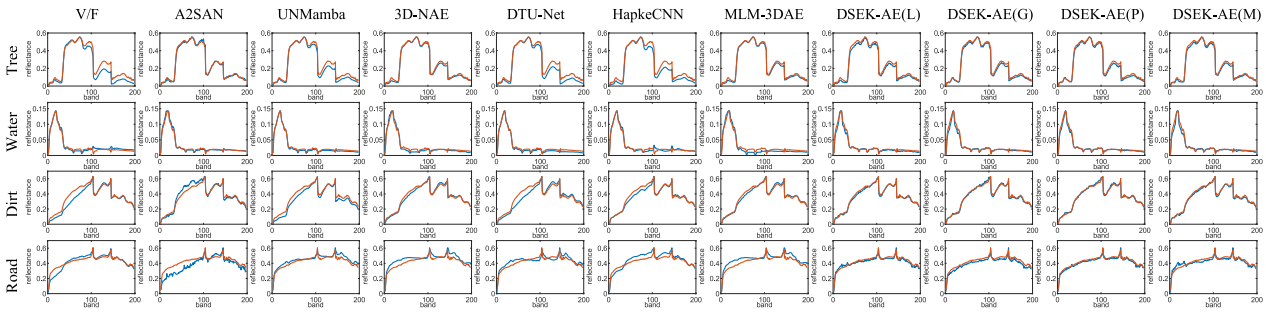


Fig. 4. Comparison of endmembers extracted by different methods (blue line) with ground reference (orange line) on the Jasper Ridge dataset.

TABLE II

PERFORMANCE EVALUATION (MEAN ± STD) OF DIFFERENT METHODS ON THE JASPER RIDGE DATASET

Methods	V/F [5] [6]	A2SAN [49] (2024)	UNMamba [29] (2025)	3D-NAE [31] (2022)	DTU-Net [42] (2026)	HapkeCNN [43] (2022)	MLM-3DAE [44] (2024)	DSEK-AE			
								LMM	GBM	PPNM	MLM
aSAD	0.1069 ±0.0186	0.0767 ±0.0055	0.0695 ±0.0049	0.0730 ±0.0066	0.0996 ±0.0018	0.0868 ±0.0042	0.0785 ±0.0145	0.0492 ±0.0019	0.0453 ±0.0037	**0.0364 ±0.0037**	0.0537 ±0.0032
aRMSE	0.1598 ±0.0360	0.1054 ±0.0084	0.0698 ±0.0058	0.1336 ±0.0053	0.0904 ±0.0022	0.1370 ±0.0090	0.0865 ±0.0229	0.0634 ±0.0029	0.0619 ±0.0051	**0.0578 ±0.0015**	0.0664 ±0.0058

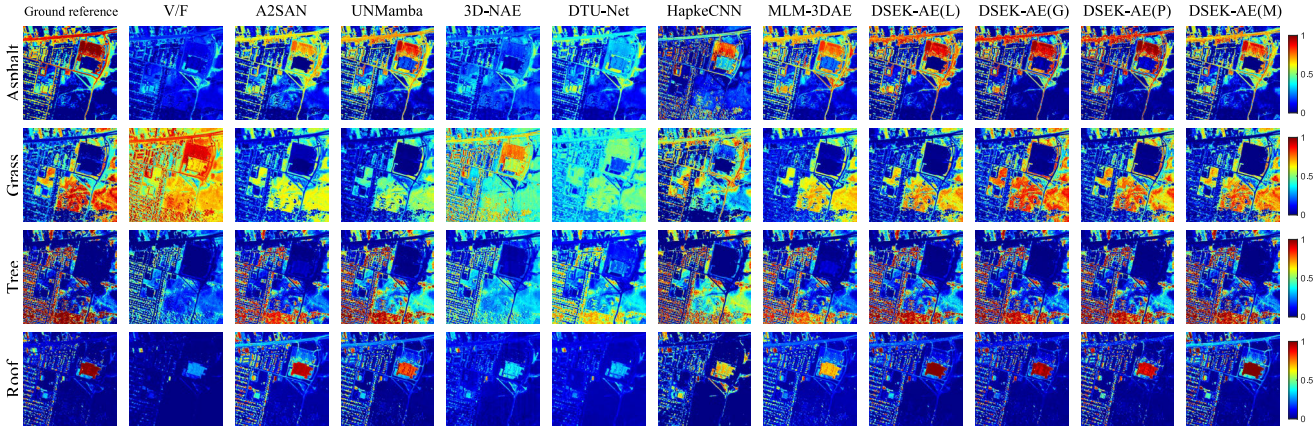


Fig. 5. Abundance maps obtained by different methods for the Urban dataset.

order scattering effects, it is specifically tailored for intimate mixed scenarios, which differ from the actual condition of the Orchard dataset; consequently, its performance in endmember extraction is not satisfactory. The poor result of 3D-NAE in abundance estimation may be attributed to its reliance on learning the spectral mixing patterns from the data; given

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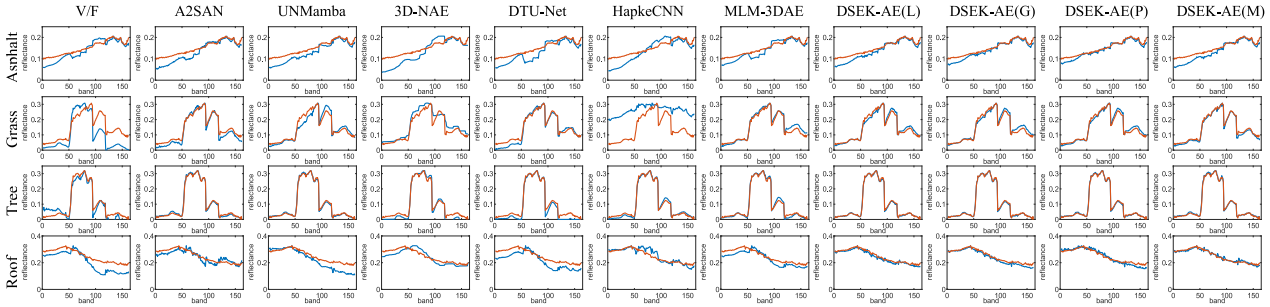


Fig. 6. Comparison of endmembers extracted by different methods (blue line) with ground reference (orange line) on the Urban dataset.

TABLE III

PERFORMANCE EVALUATION (MEAN ± STD) OF DIFFERENT METHODS ON THE URBAN DATASET

Methods	V/F [5] [6]	A2SAN [49] (2024)	UNMamba [29] (2025)	3D-NAE [31] (2022)	DTU-Net [42] (2026)	HapkeCNN [43] (2022)	MLM-3DAE [44] (2024)	DSEK-AE			
								LMM	GBM	PPNM	MLM
aSAD	0.2117 ±0.0205	0.0958 ±0.0176	0.1171 ±0.0075	0.1429 ±0.0092	0.1309 ±0.0018	0.1792 ±0.0085	0.1094 ±0.0141	0.0587 ±0.0074	**0.0548 ±0.0022**	0.0553 ±0.0023	0.0582 ±0.0052
aRMSE	0.2930 ±0.0112	0.1253 ±0.0123	0.1156 ±0.0062	0.2657 ±0.0023	0.2190 ±0.0094	0.2585 ±0.0077	0.1330 ±0.0114	0.0975 ±0.0042	**0.0889 ±0.0032**	0.0934 ±0.0052	0.0986 ±0.0052

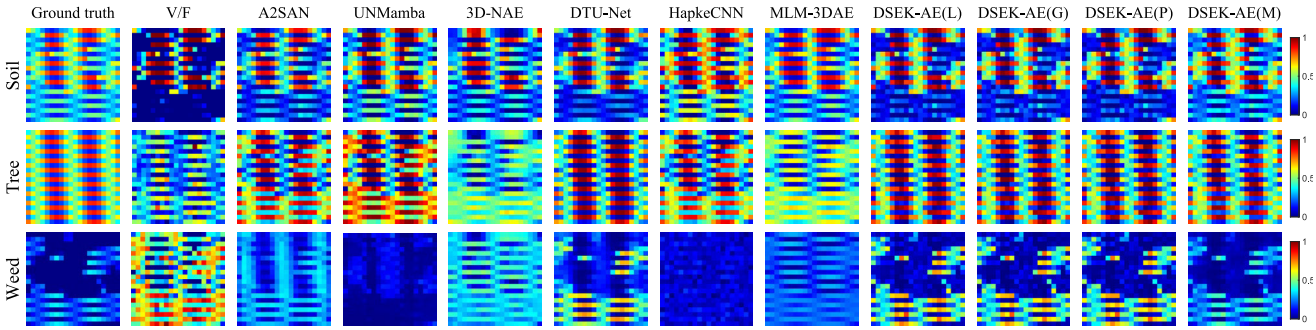


Fig. 7. Abundance maps obtained by different methods for the Orchard dataset.

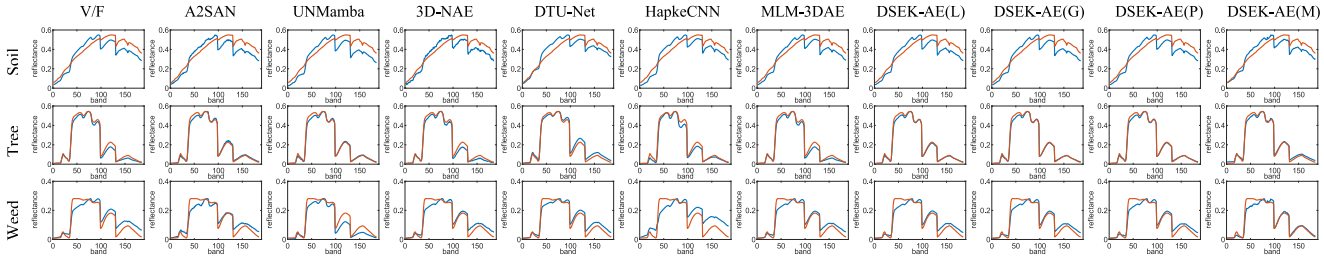


Fig. 8. Comparison of endmembers extracted by different methods (blue line) with ground truth (orange line) on the Orchard dataset.

TABLE IV

PERFORMANCE EVALUATION (MEAN ± STD) OF DIFFERENT METHODS ON THE ORCHARD DATASET

Methods	V/F [5] [6]	A2SAN [49] (2024)	UNMamba [29] (2025)	3D-NAE [31] (2022)	DTU-Net [42] (2026)	HapkeCNN [43] (2022)	MLM-3DAE [44] (2024)	DSEK-AE			
								LMM	GBM	PPNM	MLM
aSAD	0.1332 ±0.0096	0.1295 ±0.0073	0.1260 ±0.0013	0.1269 ±0.0089	0.1189 ±0.0202	0.1439 ±0.0007	0.1259 ±0.0117	0.1174 ±0.0015	0.1178 ±0.0024	0.1166 ±0.0014	**0.1032 ±0.0024**
aRMSE	0.2758 ±0.0030	0.1524 ±0.0106	0.1538 ±0.0031	0.2243 ±0.0042	0.1398 ±0.0144	0.1307 ±0.0001	0.1343 ±0.0090	0.1481 ±0.0021	0.1423 ±0.0042	0.1467 ±0.0038	**0.0838 ±0.0041**

the insufficient number of pixels in the Orchard dataset, it is challenging to learn an accurate mixing model from it. Figs. 7 and 8 present the visualization results of abundance estimation and endmember extraction, respectively, which are consistent

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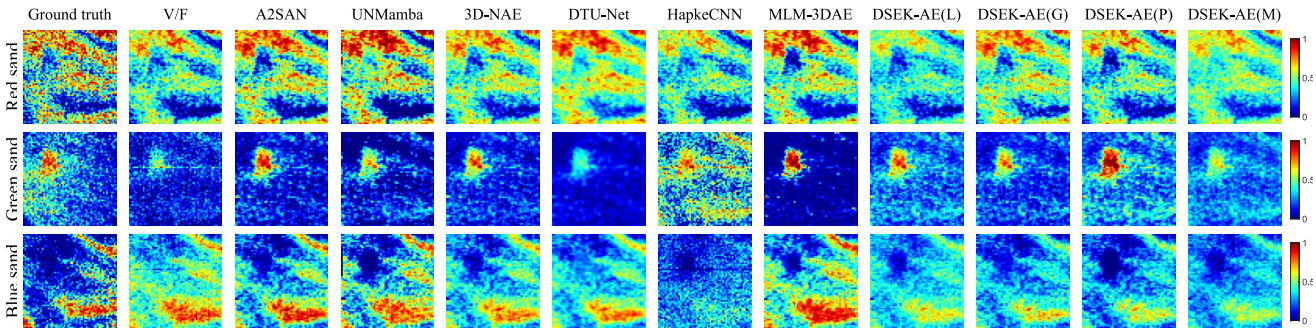


Fig. 9. Abundance maps obtained by different methods for the laboratory-created dataset.

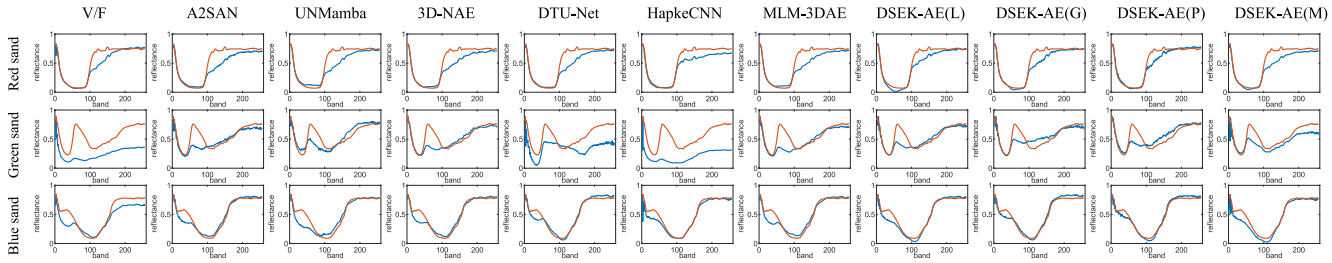


Fig. 10. Comparison of endmembers extracted by different methods (blue line) with ground truth (orange line) on the laboratory-created dataset.

TABLE V
PERFORMANCE EVALUATION (MEAN ± STD) OF DIFFERENT METHODS ON THE LABORATORY-CREATED DATASET

Methods	V/F [5] [6]	A2SAN [49] (2024)	UNMamba [29] (2025)	3D-NAE [31] (2022)	DTU-Net [42] (2026)	HapkeCNN [43] (2022)	MLM-3DAE [44] (2024)	DSEK-AE			
								LMM	GBM	PPNM	MLM
aSAD	0.1816 ±0.0039	0.1660 ±0.0085	0.1831 ±0.0051	0.1698 ±0.0048	0.1638 ±0.0079	0.1583 ±0.0029	0.1813 ±0.0036	0.1339 ±0.0023	0.1287 ±0.0058	0.1252 ±0.0048	**0.1194 ±0.0020**
aRMSE	0.2618 ±0.0153	0.2362 ±0.0008	0.2576 ±0.0029	0.2250 ±0.0040	0.2355 ±0.0060	0.2489 ±0.0012	0.2749 ±0.0218	0.2205 ±0.0047	0.2191 ±0.0034	0.2196 ±0.0039	**0.2089 ±0.0013**

with the findings in Table IV. The results of the proposed method with the MLM kernel are closest to the ground truth.

4) Experiments on the Laboratory-Created Dataset: With complex mixture patterns, spatially irregular sand distribution, and limited data volume, the laboratory-created dataset poses significant challenges for unmixing. Consequently, the performance of all methods on this dataset is not particularly satisfactory. In comparison, our method yields superior performance in both endmember extraction and abundance estimation. Table V lists the aSAD and aRMSE obtained on the laboratory-created dataset. Because of the high degree of mixing in the laboratory-created dataset, the absence of pure pixels makes it challenging to obtain good estimates of the initial values of the endmembers. As a result, the methods that rely on good initialization fail to produce accurate endmember extraction results. In contrast, our methods leverage endmember prior and image information to learn endmembers directly from the data, thereby achieving better results. The abundance maps and endmembers estimated by each method are depicted in Figs. 9 and 10, respectively. It can be seen that the proposed method with the MLM kernel corresponds most closely to the ground truth.

D. Ablation Experiments

1) Investigation on Endmember Initialization Sensitivity: The results of most unmixing algorithms heavily rely on the initialization of endmembers. In the above experiments, relatively favorable initial values are used to enhance the performance of the comparative methods. To better demonstrate the advantage of the proposed method in directly estimating endmembers, Fig. 11 presents the endmember results extracted from all methods under poor initialization conditions on the Jasper Ridge dataset. To ensure fairness, our method also employs this same poor initial value as guidance.

As illustrated in Fig. 11, due to the poor initial value, the methods that extract endmembers from the weights of a specific layer within the decoder yield inaccurate results, closely resembling their initial value. In contrast, A2SAN, UNMamba, and the proposed method, which directly compute endmembers, achieve more satisfactory results, and our method demonstrates a closer alignment with the ground reference. To enhance the reliability of the conclusions, we also conducted experiments on the Orchard dataset. As shown in Fig. 12, the experimental results on the Orchard dataset yield conclusions highly consistent with those obtained from the Jasper Ridge dataset.

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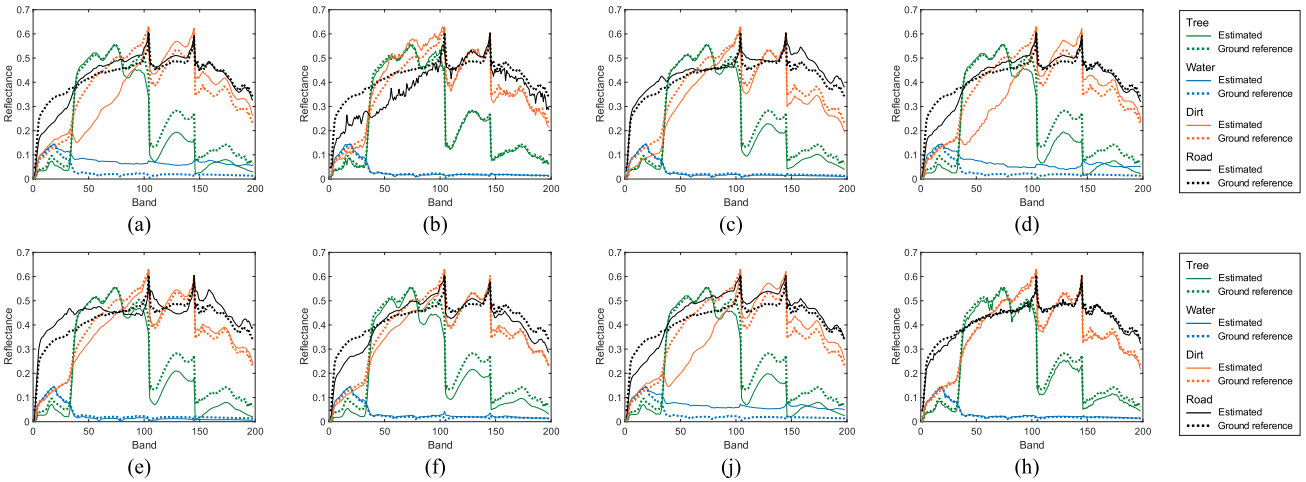


Fig. 11. Endmembers obtained by different methods under poor initialization/guidance conditions on the Jasper Ridge dataset. (a) Initialization. (b) A2SAN. (c) UNMamba. (d) 3D-NAE. (e) DTU-Net. (f) HapkeCNN. (j) MLM-3DAE. (h) DSEK-AE(P).

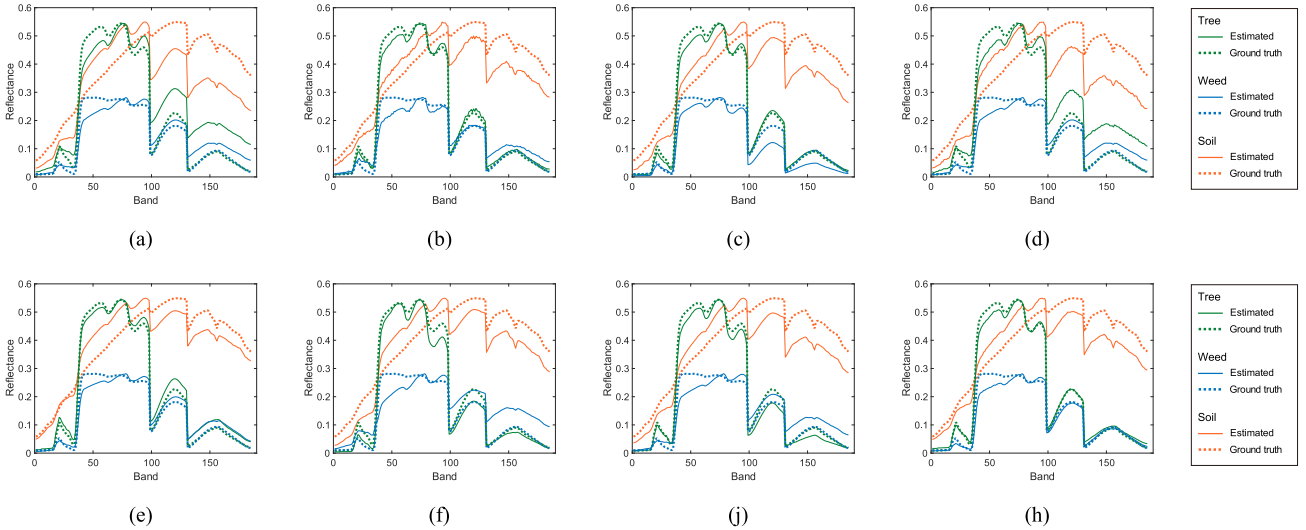


Fig. 12. Endmembers obtained by different methods under poor initialization/guidance conditions on the Orchard dataset. (a) Initialization. (b) A2SAN. (c) UNMamba. (d) 3D-NAE. (e) DTU-Net. (f) HapkeCNN. (j) MLM-3DAE. (h) DSEK-AE(M).

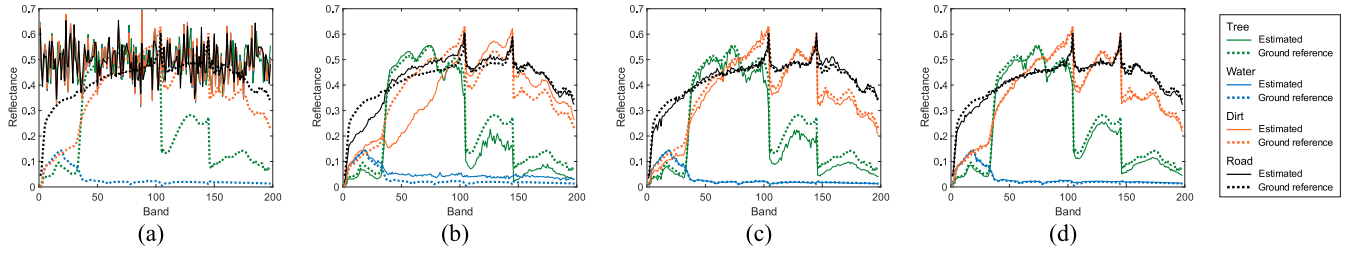


Fig. 13. Learning process of endmembers. (a) Epoch 0. (b) Epoch 80. (c) Epoch 160. (d) Epoch 240.

To provide a more detailed demonstration of the insensitivity of DSEK-AE to this poor endmember guidance, the learning process of the endmembers on the Jasper Ridge dataset is shown in Fig. 13. Initially, endmembers are random and disordered. Under the guidance of the initial estimates, they are gradually optimized and then further refined in the second training stage to yield the final results of endmembers. By comparing Figs. 11(a) with 13(d), it can be observed that the final endmember results exhibit substantial improvement over the guidance and show high similarity to the ground reference.

Moreover, Fig. 14 shows that the training loss and aSAD gradually decreases and eventually stabilize with increasing training iterations, which is consistent with the evolution of

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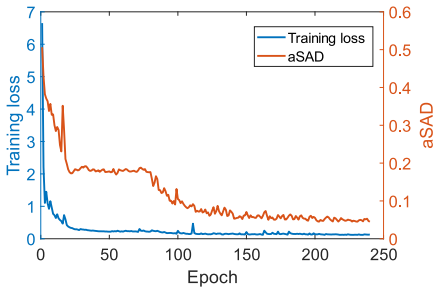


Fig. 14. Convergence curve.

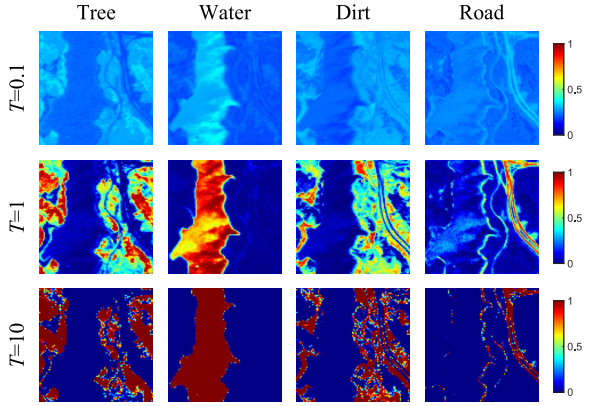


Fig. 15. Abundance output of the SAL across varying T.

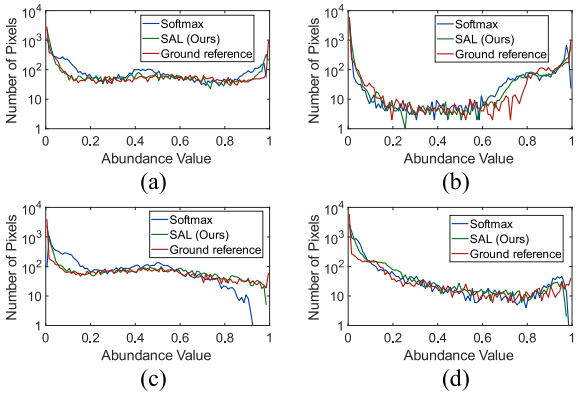


Fig. 16. Histograms of abundance maps for each endmember on the Jasper Ridge dataset. (a) Tree. (b) Water. (c) Dirt. (d) Road.

endmembers observed in Fig. 13, providing evidence for the convergence of the proposed method.

2) Analysis of the Effect of SAL: To better adjust the sparsity of abundances during training according to input, SAL is introduced, which adaptively regulates the sparsity of abundances by learning an optimal T.

To more intuitively understand the impact of SAL, Fig. 15 shows the output results of SAL on the Jasper Ridge dataset when the input remains unchanged but the value of T varies. Clearly, as T increases, the abundances become increasingly sparse. To further analyze the role of SAL, we replace SAL with softmax, sparsemax, and entmax ($\alpha = 1.5$) functions [62] respectively to satisfy the abundance's ASC and ANC while keeping the rest of the network unchanged. The unmixing performance on the Jasper Ridge dataset using the PPNM kernel is shown in Table VI. The results indicate that SAL outperforms other functions. Furthermore, we plot the histograms of abundance maps obtained using SAL and the softmax function. As shown in Fig. 16, the results obtained by our method are closer to the ground reference than those obtained using the softmax function, which is consistent with the aRMSE results in Table VI, further demonstrating the effectiveness of SAL.

Fig. 17 shows the variation of T during the training process, which eventually approaches a relatively stable value.

3) Ablation Study of Network Components: To investigate the impact of network components on the performance of the proposed method, we conduct the following ablation studies.

TABLE VI
PERFORMANCE EVALUATION OF DIFFERENT NORMALIZATION METHODS

Normalization Methods	Softmax	Sparsemax	Entmax	Ours
aSAD	0.0566	0.0515	0.0503	**0.0368**
aRMSE	0.0714	0.0744	0.0678	**0.0577**

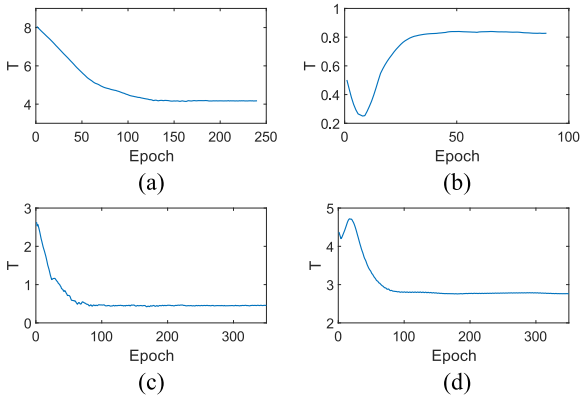


Fig. 17. Variation of T during training. (a) Jasper Ridge, (b) Urban, (c) Orchard, and (d) Laboratory-created datasets.

TABLE VII
ABLATION STUDY ON THE ENDMEMBER EXTRACTION STREAM

Methods	aSAD	aRMSE
Endmember extraction stream → 1×1 CNN	0.0644	0.0778
FCL → CNN (followed by average pooling)	0.0728	0.0847
Ours	**0.0368**	**0.0577**

TABLE VIII
ABLATION STUDY ON GLAM

Methods	aSAD	aRMSE
DSEK-AE (w/o transformer branch)	0.0591	0.0810
DSEK-AE (w/o CNN branch)	0.0560	0.0782
Ours	**0.0368**	**0.0577**

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TABLE IX
PERFORMANCE EVALUATION OF DIFFERENT LOSS FUNCTIONS

L_{SAM}	L_{ED}	L_M	Two-stage	L_N	Metrics	
					aSAD	aRMSE
√	×	×	×	×	0.1057	0.1111
√	√	×	×	×	0.0965	0.0922
√	√	√	×	×	0.0717	0.0894
√	√	√	√	×	0.0555	0.0637
√	√	√	×	√	0.0649	0.0654
√	√	√	√	√	**0.0368**	**0.0577**

First, we remove the endmember extraction stream and introduce a 1×1 convolutional layer in the decoder, using its weights as endmembers, similar to current mainstream methods. Second, to verify the effectiveness of FCL, we replaced it with CNN (followed by average pooling). The experimental results on the Jasper Ridge dataset using the PPNM kernel are shown in Table VII, demonstrating the effectiveness of the endmember extraction stream in improving unmixing accuracy and highlighting the superiority of FCL over CNN in endmember extraction. Besides, we perform ablation studies on the upper and lower branches of GLAM, with the results on the Jasper Ridge dataset using the PPNM kernel presented in Table VIII, verifying the effectiveness of combining CNN with transformer.

4) Ablation Study of Loss Function: To validate the necessity and effectiveness of each term in the loss function, ablation experiments are conducted.

The results of training the network with different loss functions on the Jasper Ridge dataset using the PPNM kernel are presented in Table IX. It can be concluded that evaluating the reconstruction error from multiple perspectives, employing a two-stage training process, and applying regularizations on both endmembers and nonlinear intensities all contribute to improving unmixing accuracy.

5) Analysis of Hyperparameter Settings: To analyze the impact of the hyperparameters in the loss function on the unmixing results and determine their values, the aSAD and aRMSE obtained by the proposed method under different hyperparameter settings on the four datasets are shown in Table X, with the best results marked in bold.

Overall, the unmixing results are not sensitive to changes in hyperparameters within a certain range. Therefore, in the experiments, λ_{ED}, λ_M, and λ_N are set to 10^{-2}, 1, and 10^{-1} except for the laboratory-created dataset, where they are set to 10^{-5}, 10^{-3}, and 10^{-4}, respectively. In addition, the edge density of an image is calculated by dividing the total number of edge pixels detected using the Canny operator by the total number of image pixels and is used to characterize the spatial smoothness of the image. As shown in Table X, the edge density of the laboratory-created dataset is significantly higher than that of the other three datasets, reflecting significantly lower spatial smoothness in the laboratory-created dataset, which is consistent with visual observations. Thus, we speculate that the difference in hyperparameter settings between these two types of datasets may be due to their differing spatial structural characteristics.

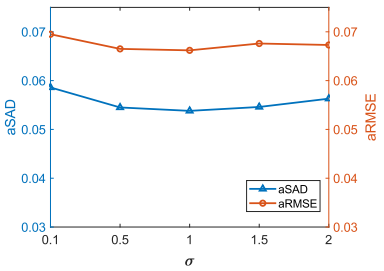


Fig. 18. Results under different σ.

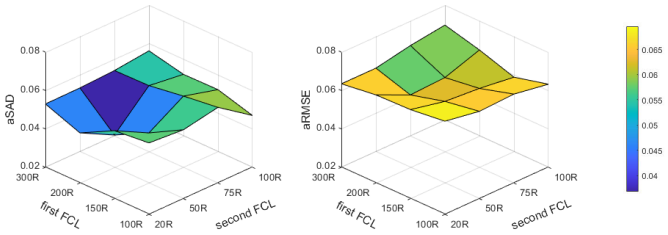


Fig. 19. Results under different hidden layer dimensions.

Furthermore, we conduct a sensitivity analysis of Gaussian kernel parameter σ on the Jasper Ridge dataset. As shown in Fig. 18, the experimental results indicate that the performance of the proposed method remains relatively stable within a certain range of σ. Considering that $\sigma = 1$ yields the best result in the sensitivity experiment analysis, and from a mathematical perspective, the Gaussian kernel parameter σ acts as a "distance scale," and $\sigma = 1$ matches the order of magnitude of the normalized input [63], therefore, σ is set to 1 in all other experiments.

To investigate the impact of different hidden layer dimensions in the endmember extraction stream on unmixing accuracy, we also conduct related experiments, the results of which are shown in Fig. 19. As seen from Fig. 19, although performance degrades when the hidden layer dimension is either too large or too small, the magnitude of variation remains relatively small. Overall, within a certain range, the model shows relatively low sensitivity to changes in hidden layer dimension.

6) Complexity Analysis: In the experiments, all methods are run on a computer with an Intel[1] Core[2] i9-9900K CPU @ 3.60 GHz and an NVIDIA GeForce RTX3090 GPU. The complexity of the algorithm is evaluated in terms of the running time and the number of model parameters, with the results of different methods presented in Table XI. To comprehensively and effectively extract spatial-spectral information from HSIs, the proposed DSEK-AE employs transformers for abundance estimation. Because of the larger number of parameters and higher computational complexity of transformers compared to CNNs, the proposed method exhibits a relatively large parameter count (as shown in Table XI). However, compared

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[2]Trademarked.

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TABLE X
RESULTS UNDER DIFFERENT HYPERPARAMETER SETTINGS

Hyperparameter Settings		λ_{ED}					λ_M					λ_N					Edge Density
		10^{-4}	10^{-3}	10^{-2}	10^{-1}	1	0.1	0.5	1	2	5	0.01	0.05	0.1	0.5	1	
Jasper	aSAD	0.0573	0.0469	**0.0368**	0.0462	0.0552	0.0646	0.0535	**0.0368**	0.0494	0.0668	0.0574	0.0478	**0.0368**	0.0456	0.0558	0.1260
	aRMSE	0.0791	0.0652	**0.0577**	0.0590	0.0695	0.0795	0.0755	**0.0577**	0.0699	0.0917	0.0799	0.0785	**0.0577**	0.0691	0.0758	
Urban	aSAD	0.0635	0.0664	**0.0554**	0.0628	0.0558	0.0667	0.0585	**0.0554**	0.0555	0.0670	0.0684	0.0555	**0.0554**	0.0660	0.0581	0.1467
	aRMSE	0.0909	0.0955	**0.0889**	0.0911	0.0895	0.1118	0.0993	0.0889	**0.0854**	0.0954	0.1219	**0.0837**	0.0889	0.1094	0.1264	
Orchard	aSAD	0.1035	0.1038	**0.1030**	0.1065	0.1097	0.0991	**0.0986**	0.1030	0.1171	0.1226	0.1088	0.1064	**0.1030**	0.1071	0.1153	0.1461
	aRMSE	0.1316	0.0949	**0.0839**	0.0866	0.0879	0.0918	0.1135	**0.0839**	0.1320	0.1384	0.0879	0.0863	**0.0839**	0.0873	0.0984	
Hyperparameter Settings		λ_{ED}					λ_M					λ_N					Edge Density
		10^{-7}	10^{-6}	10^{-5}	10^{-4}	10^{-3}	0.0001	0.0005	0.001	0.005	0.01	10^{-6}	10^{-5}	10^{-4}	10^{-3}	10^{-2}	
Laboratory-created	aSAD	0.1275	0.1211	**0.1198**	0.1256	0.1347	0.1322	0.1236	**0.1198**	0.1225	0.1311	0.1309	0.1277	**0.1198**	0.1253	0.1317	0.2241
	aRMSE	0.2185	0.2099	**0.2083**	0.2118	0.2244	0.2192	0.2141	**0.2083**	0.2172	0.2175	0.2192	0.2188	**0.2083**	0.2110	0.2240	

TABLE XI
RUNNING TIME AND THE NUMBER OF MODEL PARAMETERS OF DIFFERENT METHODS

Methods		V/F	A2SAN	UNMamba	3D-NAE	DTU-Net	HapkeCNN	MLM-3DAE	DSEK-AE
Jasper	Time (s)	1.2613	15.9941	50.4756	148.4386	39.6528	14.8682	402.8047	22.3300
	Params (M)	—	0.04	0.23	0.64	10.90	1.65	0.13	13.06
Urban	Time (s)	4.5370	46.0525	68.1932	480.7200	173.8275	38.7427	2200.2573	93.7748
	Params (M)	—	0.04	2.98	0.56	118.91	1.57	0.10	132.69
Orchard	Time (s)	0.1153	10.9080	39.3651	5.2446	48.4798	8.7057	20.8718	14.3781
	Params (M)	—	0.04	0.15	0.57	18.22	1.62	0.11	1.77

TABLE XII
PERFORMANCE EVALUATION (MEAN ± STD) OF DIFFERENT METHODS WITH VARYING SNR

SNR	Metrics	V/F	A2SAN	UNMamba	3D-NAE	DTU-Net	HapkeCNN	MLM-3DAE	DSEK-AE(P)
15dB	aSAD	0.1607±0.0061	0.1528±0.0081	0.0973±0.0304	0.1259±0.0079	0.1462±0.0114	0.1554±0.0036	0.1228±0.0049	**0.0728±0.0041**
	aRMSE	0.1687±0.0105	0.1367±0.0090	0.0845±0.0189	0.1711±0.0112	0.1239±0.0079	0.1608±0.0088	0.1471±0.0068	**0.0759±0.0011**
20dB	aSAD	0.1352±0.0050	0.1329±0.0148	0.0856±0.0182	0.1126±0.0081	0.1368±0.0103	0.1473±0.0036	0.1076±0.0040	**0.0652±0.0041**
	aRMSE	0.1636±0.0021	0.1307±0.0150	0.0784±0.0107	0.1470±0.0039	0.1057±0.0145	0.1458±0.0168	0.1463±0.0073	**0.0686±0.0033**
25dB	aSAD	0.1282±0.0049	0.1223±0.0070	0.0781±0.0183	0.0924±0.0167	0.1119±0.0083	0.1043±0.0157	0.0902±0.0098	**0.0538±0.0037**
	aRMSE	0.1608±0.0023	0.1235±0.0130	0.0742±0.0192	0.1409±0.0050	0.0964±0.0122	0.1397±0.0030	0.1223±0.0142	**0.0617±0.0017**

to other AE-based methods, our method still maintains a relatively short runtime.

In addition, it should be noted that although transformers typically have relatively large network sizes and a high number of parameters, the HSI datasets used in the experiments contain a large number of spectral bands, thereby providing a sufficiently large data volume for training.

7) Analysis of Noise Robustness: HU aims to accurately retrieve endmembers and their abundances from the acquired data. Usually, high-quality hyperspectral data are required, and the unmixing accuracy is also relatively sensitive to noise levels. Under conditions of severe data missing or heavy noise contamination, effective HU is usually difficult to achieve. Next, we examine the noise robustness of the proposed method.

Specifically, to verify the noise robustness of the proposed method, Gaussian noise with signal-to-noise ratio (SNR) levels of 15, 20, and 25 dB is added to the Jasper Ridge dataset. The hyperparameter settings are kept consistent with the previous configuration. To mitigate the impact of randomness, repeated experiments are carried out for each noise level, with Gaussian noise independently and randomly added in each trial. Table XII presents the experimental results on the noisy datasets.

Because of the noise, all methods show a performance drop. Furthermore, as the noise increases, the results of all methods in endmember extraction and abundance estimation gradually deteriorate. However, the proposed method still performs the best, demonstrating good noise robustness.

E. Discussion

Based on the above experimental results, the insightful analysis is given as follows.

1) Unmixing Accuracy: Based on the encoder-decoder architecture, the proposed DSEK-AE achieves high-precision blind HU by fully leveraging image information and inte-

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grating the physical (a priori) knowledge into the unmixing network.

Specifically, the endmember extraction stream consolidates all pixel information and achieves spectral smoothness of endmembers through 1D-CNNs, enabling precise endmember computation; the abundance estimation stream uses the CNN-transformer hybrid network to effectively extracts the spatial-spectral information of the input HSIs, captures the spatial smoothness of abundances by using convolutional layers with a kernel size of 3×3, and dynamically adjusts the abundance sparsity with the subsequent SAL, thereby enhancing abundance estimation accuracy; the endmember-kernel theory is incorporated into the decoder, allowing it to flexibly represent and interpret various spectral mixing scenarios.

It is due to the substantial physical knowledge fully integrated into the network architecture design, realizing a dual-driven approach of knowledge and data, that the proposed DSEK-AE combines the advantages of model-based methods and deep-learning methods, thus possessing strong physical interpretability and achieving high accuracy in unmixing results.

Besides, to effectively address the problem of blind HU, a composite loss function is designed, which further enables the proposed DSEK-AE to exhibit excellent performance across all datasets.

2) Complexity Analysis: The proposed method significantly enhances unmixing accuracy and boasts relatively shorter running time compared to other AE-based approaches, which should hold important potential value for practical applications in HSIs. However, due to the introduction of the transformer modules, the proposed DSEK-AE still contains a large number of parameters, which may limit its deployment in resource-constrained scenarios. In future work, we will explore lightweight network architecture designs—such as replacing the full self-attention mechanism with more efficient attention mechanisms or substituting the transformer module with lighter alternatives like Mamba—to reduce computational cost and memory consumption while maintaining high unmixing accuracy.

3) Flexibility for Different Spectral Mixing Scenarios: Thanks to the endmember-kernel theory, the proposed DSEK-AE can easily switch the embedded spectral mixing model by replacing the kernel matrix without modifying the basic architecture of the unmixing network. Therefore, it can conveniently embed different spectral mixing models according to different application scenarios.

The experimental results on the four datasets with varying spectral mixing characteristics indicate that using an appropriate kernel yields optimal performance, whereas kernel mismatch results in performance degradation. This demonstrates that by flexibly adjusting the kernel type, the network can better handle various spectral mixing scenarios, significantly enhancing its flexibility and adaptability across different scenarios.

4) Insensitivity to Endmember Initialization: Considering the sensitivity of endmembers to initialization, the endmember extraction stream uses FCLs and 1D-CNNs to directly compute endmembers from the observed data. This design integrates all information from the original input while considering endmember priors, enhancing the accuracy of endmember extraction.

Since blind unmixing is an ill-posed inverse problem belonging to nonconvex optimization, it typically involves numerous local optima, and completely random initialization without physical meaning often results in poor results. Therefore, in the first stage of training, the results from VCA are used to guide the network in learning endmembers. Although VCA results are generally not precise, they provide guidance for the learning of endmember spectral curves with physical meaning. In our method, other automatic endmember extraction approaches (such as SGA, OBA, etc.) can also be used to achieve this purpose/guidance. During the second stage of training, the network moves beyond the influence of the initial endmember guidance and further learns endmember spectral curves that are closer to real ones.

Through the endmember extraction stream and the two-stage training strategy, the network ultimately acquired the endmember spectral curves closest to the real endmembers, significantly reducing the network's sensitivity to endmember initialization. The experimental results also verified this conclusion.

5) Robustness to Noise: In SPM, the input hyperspectral imagery undergoes dimensionality reduction, which helps the network focus on the intrinsic structure and features of the data during the learning process, thereby effectively suppressing noise and irrelevant information. Furthermore, the proposed DSEK-AE comprehensively integrates the physical (a priori) knowledge and specific spectral mixing models into the network architecture design and imposes regularizations on endmembers and nonlinear intensities in the loss function, thereby restricting the feasible solution space of the network, suppressing overfitting to noise, and consequently reducing the network's sensitivity to noise. The experimental results show that the proposed DSEK-AE achieves favorable performance across varying noise levels.

6) Hyperparameter Setting: In the above experiments, the Jasper Ridge, Urban, and Orchard datasets share the same set of hyperparameters, while the laboratory-created dataset requires a unique set of hyperparameters to achieve satisfactory unmixing results. This discrepancy may, to some extent, limit the universality of the proposed method in practical application. To analyze the reasons behind this phenomenon, after calculating the edge densities of these four datasets to characterize their spatial smoothness, we speculate that differences in hyperparameter settings are likely related to the spatial smoothness of these datasets. In future work, we will conduct a more comprehensive multifactor analysis to better understand the underlying causes and provide guidance for hyperparameter settings.

V. Conclusion

In this article, a blind linear/nonlinear HU AE with a dual-stream encoder and a decoder based on endmember kernel has been proposed. The encoder directly computes endmembers from HSIs, significantly reducing the impact of endmember initialization, while leveraging a CNN-transformer hybrid network combined with SAL to capture both global and local spatial-spectral information and achieve adaptive regulation

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of abundance sparsity. The decoder enables flexible switching between embedded spectral mixing models by simply replacing the endmember-kernel matrix. Moreover, the composite loss function comprehensively considers reconstruction error as well as regularization of endmembers and nonlinear intensities, effectively guiding network training. The experimental results have demonstrated that the proposed method achieves high-precision unmixing across multiple datasets. Meanwhile, it not only exhibits strong physical interpretability but also demonstrates superior noise robustness and flexible adaptability to various spectral mixing scenarios, indicating significant potential for practical applications.

Although the proposed method has several advantages, it still has some limitations, such as dependency on the choice of kernel function type, a relatively large number of network parameters, and uncertainty regarding factors influencing hyperparameter settings. Future work will explore automatic methods for selecting kernel function types, develop lightweight architecture designs, and conduct multifactor analysis to identify the true causes affecting hyperparameter settings, thereby addressing these issues.

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